

Register No. :.....

Name:.....

	Govt. Model Engineering College			
	Question Paper	CSBS-2024 scheme		
Computer Science and Business Systems		Computer Engineering		
CU32024-2028 Batch odd s3 2025-2026		Academic Year: 2025-26		
Course Code: PCCST303	L-T-P Credits: 3-1-0-4	Course Name: DATA STRUCTURES AND ALGORITHMS		
Time: 1.5 Hrs Hrs.	Tutorial 1	Date: 6-8-2025	Maximum Marks: 20	

No.	Question	Mark	CO
1	1. Find upper bound for $f(n) = 3n + 8$	3.0	CO1
2	1. Find upper bound for $f(n) = 2n^3 - 2n^2$	3.0	CO1
3	1. Find lower bound for $f(n) = 5n^2$	3.0	CO1
4	Find Θ bound for $f(n) = \frac{n^2}{2} - \frac{n}{2}$	3.0	CO1
5	Find the complexity of the function given below. void function(int n) { int i, count =0; for(i=1; i*i<=n; i++) count++; }	4.0	CO1
6	What is the complexity of the program given below: void function(int n) { int i, j, k , count =0; for(i=n/2; i<=n; i++) for(j=1; j + n/2<=n; j= j+1) for(k=1; k<=n; k= k * 2) count++; }	4.0	CO1